Policy Learning with Surrogate Approximators for Block-Scale Urban Morphology Design: A Comparative Benchmarking Study

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Abstract

Integrating urban morphology generation, performance prediction, and design search is essential for low-carbon urban design, yet full simulation-based optimization remains computationally demanding. This study develops and evaluates a surrogate-assisted deep reinforcement learning (DRL) workflow for block-scale urban morphology design under surrogate-predicted energy indicators. A deep neural network (DNN) surrogate predicts Energy Use Intensity (EUIt), Energy Generation (EG), and Sunlight Hours (H) from 12 morphological factors, and a Deep Deterministic Policy Gradient (DDPG) agent searches the design space under different reward preferences. In the revision workflow, the surrogate-training regime was expanded from 500 to 2000 simulated morphologies, and the final optimizer comparison was rerun on the strict highest-accuracy surrogate selected from that scale study. On this accuracy-first comparison, fair-budget NSGA-II attains stronger mean objective summaries, the largest Hypervolume, and the lowest IGD, while all three DDPG scenarios remain weaker under post-hoc preference-aligned utility selection. The reward settings nevertheless still yield interpretable morphology-level differences, especially in how openness, density, and solar-access descriptors shift across the three DDPG scenarios. Overall, the proposed framework is better interpreted as a preference-conditioned region-finding and design-support tool than as a universally dominant replacement for Pareto-based evolutionary search.

Keywords: Multi-objective optimization, Urban morphology, Energy performance, Deep reinforcement learning

Abbreviations and Symbols

Abbreviations

UM	Urban morphology
EP	Energy performance
UMF	Urban morphological factor
EPI	Energy performance indicator
DRL	Deep reinforcement learning
DNN	Deep neural network
DDPG	Deep Deterministic Policy Gradient
MOO	Multi-objective optimization
HV	Hypervolume
IGD	Inverted Generational Distance

Main symbols

EUIt	Energy use intensity (kWh/m^2/y)
EG	Energy generation (10^6 kWh/y)
H	Sunlight hours (h)
FAR	Floor area ratio
BD	Building density
OSR	Open space ratio
SVF	Sky view factor
AF	Average floors
OSLI	Open Space Location Index
θ	Block orientation angle (deg)
w_1, w_2, w_3	Reward weights for EUIt, EG, and H
γ, τ	DDPG discount and soft-update factors

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1. Introduction

In the context of carbon peaking and carbon neutrality, deep decarbonization of urban energy systems has become a central issue in sustainable development [1, 2]. Because cities concentrate both energy demand and renewable-energy opportunities, urban morphology (UM) has a strong influence on building energy performance (EP), solar access, and environmental quality [3, 4, 5]. However, urban design and energy planning are still often treated as separate tasks in practice, which weakens the energy performance of block-scale development and limits the effective use of renewable resources [6, 7]. Developing an integrated framework that links morphology generation, performance evaluation, and design optimization is therefore an important research problem at the intersection of architecture, urban planning, and energy systems [8].

This context also highlights the methodological importance of surrogate-based design studies. In practice, large comparative sweeps over urban-energy design strategies are often bottlenecked by expensive simulation workflows, so surrogate-assisted optimization is increasingly used as a screening layer before full physical validation. Under these conditions, optimizer reliability becomes part of the energy-design problem itself: if a learning-based method is unstable, checkpoint-sensitive, or difficult to benchmark fairly, its practical value for urban energy design is limited.

1.1. Research Motivation

To address this challenge, computational design methods based on parametric modeling and multi-objective optimization (MOO) have become mainstream [9, 10]. Evolutionary algorithms, exemplified by NSGA-II, are widely used to explore the design space and identify Pareto-optimal trade-offs among conflicting objectives such as energy consumption [11], energy generation potential [12], and daylighting [13]. However, this search-oriented paradigm frames urban morphology design primarily as Pareto-front exploration in a high-dimensional, nonlinear, and strongly coupled design space.

The core motivation of this study is therefore to test whether a preference-conditioned policy-learning formulation can identify stable high-performing surrogate regions and interpretable morphology patterns under a shared optimization budget. Rather than assuming that policy learning should dominate Pareto search, this study asks whether it offers a useful alternative inductive bias for navigating the same surrogate landscape.

1.2. Literature Review and Research Gap

The mainstream approach for multi-objective optimization in urban design relies on search-based algorithms [14, 15], particularly evolutionary algorithms such as NSGA-II [16]. These methods primarily explore complex problem spaces to identify trade-off solutions among conflicting metrics, such as energy efficiency and cost [17]. Despite their widespread adoption, this search-centric paradigm faces three fundamental limitations.

First, the computational cost is high. Evolutionary algorithms require repeated performance evaluations of large numbers of design candidates, and simulations based on engines such as EnergyPlus are inherently time-consuming [18, 19]. As design-space dimensionality increases, the time required to identify a convergent Pareto front can extend from several days to weeks [20].

Second, evolutionary algorithms inherently operate as black-box models [21]. Their search operators emphasize archive quality and diversity, but they do not directly provide a reusable policy for preference-conditioned navigation of the design space. As a result, designers are often presented with a cloud of Pareto-optimal solutions from which it is difficult to derive interpretable and transferable design strategies [22, 23].

Third, the mechanisms of NSGA-II are designed to preserve diversity and delineate the boundary of the Pareto front [24]. By contrast, a scalarized policy-learning formulation can concentrate repeated search on a narrower surrogate region. Whether such concentration reveals robust design patterns or merely amplifies surrogate bias must be tested empirically rather than assumed a priori [25].

To overcome these limitations, this study adopts Deep Reinforcement Learning (DRL) as an alternative optimization paradigm [26]. DRL combines the nonlinear feature representation capability of deep learning with the sequential decision-making framework of reinforcement learning. Instead of conducting blind searches, the agent learns a policy through repeated interaction with the environment. DRL has achieved considerable success in complex, high-dimensional decision tasks such as robotic control [27], strategy games [28], natural language processing [29], and autonomous driving [30].

These tasks share a core challenge: sequential decision-making in high-dimensional, continuous state and action spaces. This challenge mirrors urban morphology optimization, where designers must determine 12 continuously variable morphological factors. Because value-function methods such as DQN [31] are designed for discrete actions, this study focuses on the Deep Deterministic Policy Gradient (DDPG) algorithm. DDPG [32], an actor-critic method for continuous control, provides a simple and interpretable baseline for policy learning in continuous design spaces. We do not claim that DDPG is universally optimal relative to alternatives such as TD3, SAC, PPO with continuous actions, CMA-ES, MOEA/D, or Bayesian optimization. Rather, the goal of this study is to examine whether a preference-conditioned policy-learning formulation can yield useful design strategies under a shared surrogate environment.

Although DRL shows substantial promise, its application to the interdisciplinary problem of urban morphology design with surrogate energy indicators remains largely unexplored. While DRL is increasingly used for building energy control [33, 34] and energy system operation [35, 36], its use as a design-space navigation tool for urban morphology is still nascent. In particular, its behavior relative to a benchmark such as NSGA-II has not yet been rigorously characterized under a shared surrogate environment.

1.3. Study Contributions and Paper Structure

Against this background, this study develops and evaluates a DRL-DNN integrated framework for block-scale urban morphology design. Three specialist agents with distinct reward preferences are trained and compared with an NSGA-II benchmark under the same surrogate environment.

1. This study develops a DRL-DNN framework for urban morphology design by transforming the search problem into a reward-weighted policy-learning task under a shared surrogate environment.
2. This study examines whether different reward priorities lead to distinguishable search tendencies and morphology signatures within the same surrogate landscape.
3. This study compares DRL and NSGA-II under a shared surrogate environment using both Pareto-quality metrics and preference-aligned objective summaries.
4. This study combines linear correlation analysis with nonlinear response diagnostics to interpret how key UMFs influence EUIt, EG, and H.

The remainder of this paper is organized as follows. Section 2 presents the DRL-DNN integrated framework, including parametric modeling, performance simulation, surrogate modeling, and the DDPG optimization engine. Section 3 reports the experimental results and comparative analysis. Section 4 discusses the implications and limitations of the study. Section 5 concludes the paper. Additional case-setting and surrogate-validation details are provided in Appendix A, and benchmark-robustness diagnostics are reported in Appendix B.

2. Methodology

This study proposes an integrated surrogate-assisted workflow for block-scale urban morphology design. As illustrated in Fig. 1, the framework combines parametric morphology generation, performance simulation, surrogate modeling, and policy learning. A parametric dataset is first constructed to link urban morphological factors (UMFs) with key energy performance indicators (EPIs). A DNN surrogate is then trained for rapid performance prediction, after which a DDPG agent interacts with the surrogate environment to search for morphology strategies that balance energy demand, energy generation, and sunlight access.

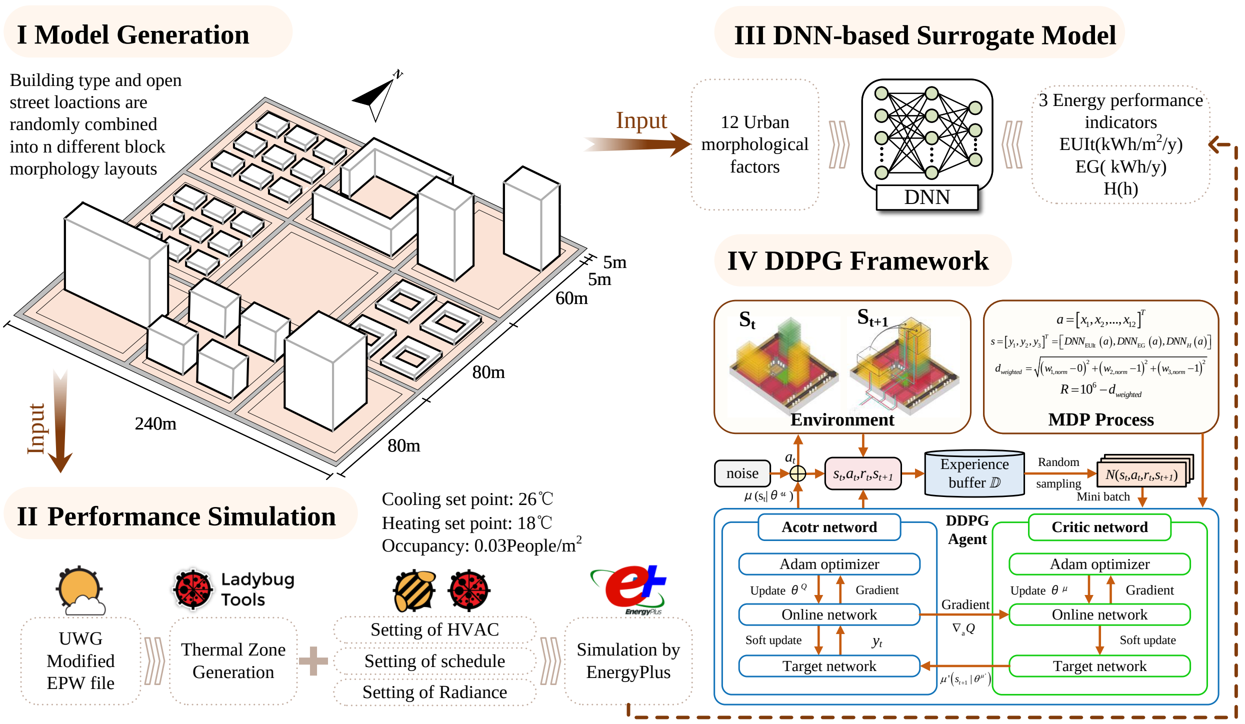

Figure 1: Overview of the proposed method. The left block summarizes parametric morphology generation and performance simulation, while the right block shows the surrogate-assisted optimization workflow used during policy learning.

2.1. Case Study and Parametric Modeling

The selected research area is Jianhu County, located in Jiangsu Province in the eastern coastal region of China. The county is situated between 33°16′ and 33°41′ N latitude and 119°33′ and 120°05′ E longitude, with an average elevation of 2 m. It has an administrative area of approximately 1160 km^2 and a population of around 600,000, resulting in a population density of about 520 persons/km^2. According to the Chinese Code for Thermal Design of Civil Buildings, Jianhu lies within the hot-summer and cold-winter climate zone and is classified as having a humid subtropical climate under the Koppen-Geiger system. The climate necessitates both building heating and cooling, as the average dry-bulb temperature is 27.1°C in the hottest month (July) and 1.6°C in the coldest month (January), with an annual average of 15.0°C. Furthermore, the region presents favorable conditions for solar energy utilization, receiving annual solar radiation of about 1250 kWh/m^2 and approximately 1800 hours of sunshine per year.

To generate a diverse dataset linking urban form to energy performance, a parametric modeling approach was implemented on the Grasshopper platform. The core of this approach involves defining a standard urban block subdivided into a 3×3 grid, creating nine land units. Based on an analysis of the existing urban fabric, a set of representative residential building prototypes, such as point, slab, and courtyard types, were defined. In the parametric model, one land unit is designated as open space, while the remaining eight units are populated with these building prototypes to generate a wide variety of block morphologies.

This parametric generation process is controlled by twelve selected UMFs. These include Floor Area Ratio (FAR), Building Density (BD), Open Space Ratio (OSR), Average Floors (AF), Setback Diversity (SD), Shape Coefficient (SC), Average Perimeter-to-Area Ratio (PAR), Aspect Ratio (AR), average Sky View Factor (SVF), Open Space Location Index (OSLI), and Block Orientation Angle θ. Once the geometric model of the block is configured on the Grasshopper platform, these UMFs can be calculated using its built-in tools. The original 500-sample surrogate regime used in the first-stage study was later expanded deterministically to 1000, 1500, and 2000 morphologies for a data-scale study, and the final optimizer comparison reported in this revision uses the strict highest-accuracy 2000-sample surrogate selected from that expanded regime. The formulas used in this work are summarized in Table 1.

Table 1: Calculation formulas of the morphological factors.

Factor	Symbol	Formula
Floor area ratio	FAR	$\mathrm{FAR} = \dfrac{\sum_{i=1}^{n} S_i}{A_s}$
Setback diversity	SD	$\mathrm{SD} = h_{\max} - h_a$
Average floors	AF	$\mathrm{AF} = \dfrac{\sum_{i=1}^{n} S_i}{\sum_{i=1}^{n} f_i}$
Aspect ratio	AR	$\mathrm{AR} = \dfrac{1}{n} \sum_{i=1}^{n} \dfrac{H_i}{W_i}$
Sky view factor	SVF	$\mathrm{SVF} = \dfrac{1}{2\pi} \int_0^{2\pi} \cos^2(\beta(R,\theta))\, d\theta$
Building density	BD	$\mathrm{BD} = \dfrac{\sum_{i=1}^{n} f_i}{A_s}$
Open space ratio	OSR	$\mathrm{OSR} = \dfrac{A_s - \sum_{i=1}^{n} f_i}{\sum_{i=1}^{n} S_i}$
Shape coefficient	SC	$\mathrm{SC} = \dfrac{\sum_{i=1}^{n} c_i}{\sum_{i=1}^{n} h_i f_i}$
Perimeter-to-area ratio	PAR	$\mathrm{PAR} = \dfrac{\sum_{i=1}^{n} p_i}{\sum_{i=1}^{n} f_i}$

Note: n is the number of buildings; OR_i is the orientation of building i; S_i is the total floor area of building i; A_s is the block area; f_i is the floor area of building i; $h_{\max}$ and h_a are the maximum and average building heights of the block; H_i and W_i describe the height and width difference of the canyon; c_i is the surface area of building i; p_i is the perimeter of building i; and β is the angle from the center point to the maximum obstacle height at a distance equal to the constant search radius R.

In addition to the factors in Table 1, OSLI is introduced to characterize the topological position of the open space within the 3×3 grid and to account for its influence on local wind, thermal, and daylight conditions [17, 18]. OSLI is an integer variable ranging from 0 to 8. Furthermore, the block orientation angle θ, ranging from $[-45°, 45°]$, is included to regulate solar access. Additional geometric details of the block layout and the representative building prototypes are provided in Appendix A.

2.2. *Performance Simulation and Optimization Objectives*

This study evaluates the energy behavior and solar potential of residential blocks using three key EPIs: total Energy Use Intensity (EUIt), Energy Generation (EG), and Sunlight Hours (H). Together, these indicators capture the demand, supply, and daylight dimensions of block-scale energy performance.

H, which indicates the duration of solar radiation exposure, was simulated using the Sunlight Hours Analysis module in Ladybug for a typical coldest day (January 20, 08:00–16:00). Continuous test points were set at 1 m intervals along south-facing ground-floor windowsills 0.9 m above ground, and the Radiance engine calculated solar trajectories and shading. The average sunlight hours across all test points represent the overall daylighting performance of the block.

EUIt, a core indicator of energy-saving performance, was simulated by constructing a Honeybee energy model based on EnergyPlus, with parameters set according to Chinese standards JGJ 134-2010 and GB 50736-2012. The simulation applied a stratified thermal zoning strategy, set the building function uniformly to residential, and used the Urban Weather Generator (UWG) to convert standard weather data into urban microclimate data to account for the urban heat island effect.

EG, representing the block's renewable-energy potential, was calculated from annual rooftop solar radiation using a coupled Ladybug-Radiance simulation. The model assumes a polycrystalline silicon PV system with 17% efficiency and 85% DC/AC conversion under standardized conditions, including 90% roof coverage and 0° tilt, to ensure comparability across cases. PV potential was then assessed against a threshold of 800 kWh/m^2/year. Finally, a Dragonfly-Ladybug toolchain was used to automate geometric information extraction and parameter transfer, thereby ensuring spatial consistency across all simulation models.

Table 2: Parameters and settings for the building energy simulation in EnergyPlus.

Classification	Parameter	Specification
System settings	Weather data	EPW files modified based on UWG urban microclimate simulations
System settings	Simulation period	Annual simulation (January 1 to December 31)
Envelope thermal performance	Wall U-value	0.8 W/(m^2·K)
Envelope thermal performance	Roof U-value	0.5 W/(m^2·K)
Envelope thermal performance	Floor U-value	1.5 W/(m^2·K)
Envelope thermal performance	Window properties	U-value = 2.70 W/(m^2·K); Solar Heat Gain Coefficient (SHGC) = 0.78
Window-to-wall ratio	North & South facades	0.4
Window-to-wall ratio	East & West facades	0.1
Internal loads	Occupant density	0.03 persons/m^2; metabolic rate = 100 W/person
Internal loads	Lighting power density	5 W/m^2
Internal loads	Equipment power density	1.9 W/m^2
Internal loads	Ventilation rate	30 m^3/(h·person)
HVAC system	Thermostat setpoints	Heating: Dec 1–Feb 28, 18°C; Cooling: Jun 15–Aug 31, 26°C
HVAC system	Infiltration rate	1 ACH (natural air leakage)

2.3. *DNN-based Surrogate Model*

In this study, a DNN is constructed as a predictive surrogate model to estimate the three target performance indicators from the simulated dataset. The model adopts a multi-layer feedforward architecture in which information flows from the input layer through the hidden layers to the output layer. Through stacked fully connected transformations, the network captures the nonlinear relationships between urban morphological factors and block-scale performance outcomes.

The DNN training process consists of two key stages: forward propagation and backpropagation. In the forward stage, the network processes input data to generate a prediction. During backpropagation, the prediction error is propagated backward from the output layer to the input layer. This process involves calculating the gradient of the loss function with respect to the network weights and iteratively updating the model parameters to reduce prediction error:

$$\mathbf{w} = \mathbf{w} + \eta \nabla_{\mathbf{w}} = \mathbf{w} + \eta \,\text{backpropagation}(f(\mathbf{x}))\,, \tag{1}$$

where η is the learning rate and $f(\mathbf{x})$ is the objective function to minimize.

To determine the DNN architecture and training settings, Bayesian Optimization (BO) was used for automated hyperparameter tuning. Compared with grid search, BO guides the search process by constructing a probabilistic model of the objective function and repeatedly selecting promising hyperparameter combinations. After the optimal settings were determined, model performance was evaluated using five-fold cross-validation. Because the surrogate is subsequently used for optimization, it is treated here as an optimization-enabling approximation rather than as a direct replacement for the original simulation workflow.

Table 3: Optimal hyperparameters of the DNN determined by Bayesian Optimization.

Hyperparameter	Value
Number of dense layers	1
Units of 1st layer	128
Units of 2nd layer	–
Dropout rate	0.143
Learning rate	0.00134
Batch size	32
Optimizer	Adam
Activation	ReLU

To ensure a rigorous evaluation of generalization capability and avoid bias from a single train-test split, the final highest-accuracy surrogate was evaluated by five-fold cross-validation over the 2000-sample regime. The coefficient of determination (R^2) and Mean Absolute Error (MAE) were used as key evaluation metrics.

Table 4: Summary of DNN prediction performance using five-fold cross-validation.

Objective	Mean R^2	Mean MAE	Std. R^2	Std. MAE
EUIt	0.983	0.209	±0.001	±0.005
EG	0.972	0.026	±0.002	±0.001
H	0.992	0.025	±0.001	±0.001

The evaluation results show that the final 2000-sample surrogate maintains R^2 values close to 1 for all three objectives while keeping MAE low relative to the observed target ranges. Relative to the best 500-sample baseline, the selected 2000-sample surrogate reduces mean target nMAE from 0.0210 to 0.0182, an improvement of about 9.9%, while also reducing mean tail nMAE from 0.0239 to 0.0203. Taken together, these results justify using the accuracy-maximized 2000-sample surrogate as the basis for the final optimizer comparison.

2.4. *DDPG-based Optimization Framework*

The parametric modeling process and the DNN surrogate together provide the basis for the integrated optimization framework linking urban morphology and energy performance. To optimize these indicators jointly, a DRL framework was developed, as shown in Fig. 2. The framework has two components. The first defines the optimization problem through the environment, action space, state representation, and reward. The second implements the DDPG agent, which interacts with the environment through an experience replay buffer and updates its policy iteratively.

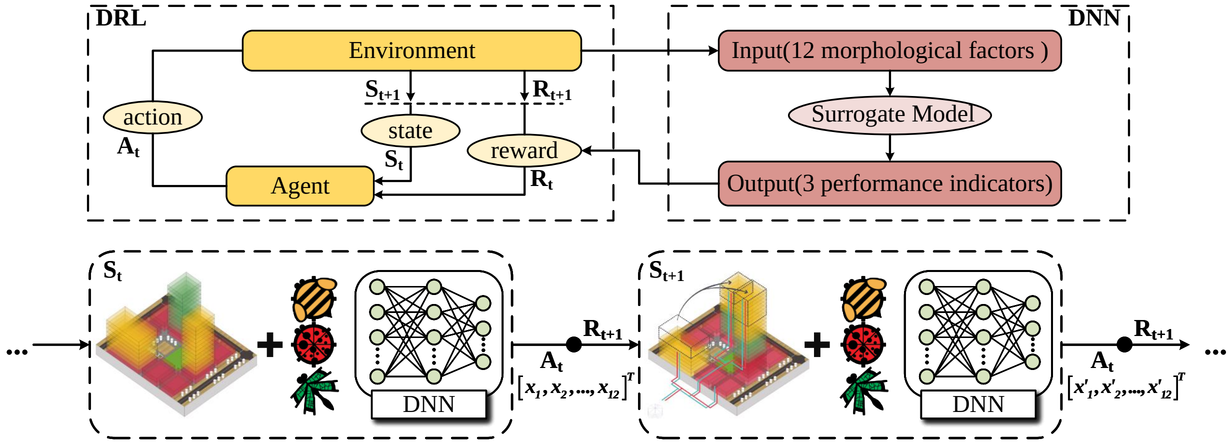

Figure 2: Conceptual framework of the DRL-based optimization. The upper part summarizes the relationship between DRL, the environment, and the DNN surrogate; the lower part details the stepwise Markov Decision Process used by the actor to query the surrogate and receive updated state and reward signals.

2.4.1. *Problem Formulation*

To address the MOO problem in urban energy performance, this study formulates the optimization process as a Markov Decision Process (MDP), typically defined by the four-tuple (S, A, P, R). The DRL agent continuously receives state and reward signals from the environment and updates its policy iteratively to improve the morphology solution.

Action space. In the DRL framework, an action is the fundamental unit of interaction between the agent and the environment. In this study, an action is defined as an adjustment to a set of preset building design parameters. Specifically, the action vector is formed by twelve key design parameters:

$$\mathbf{a} = [x_1, x_2, \ldots, x_{12}]^{\mathrm{T}}, \tag{2}$$

where $\mathbf{a}$ represents a specific action selected by the agent and x_i is the selected value for the ith design parameter.

State space. The state is defined as the vector of the key EPIs predicted by the DNN-based surrogate model:

$$\mathbf{s} = [y_1, y_2, y_3]^{\mathrm{T}} = [\mathrm{DNN}_{\mathrm{EUIt}}(\mathbf{a}), \mathrm{DNN}_{\mathrm{EG}}(\mathbf{a}), \mathrm{DNN}_{\mathrm{H}}(\mathbf{a})]^{\mathrm{T}}, \tag{3}$$

where y_1, y_2, and y_3 are the predicted values of EUIt, EG, and H, respectively.

State transition probability. The state-transition probability $P(s' \mid s, a)$ defines the probability distribution of the environment transitioning to the next state given the current state and action. In the model-free DRL method employed here, P is not modeled explicitly; instead, the agent learns directly from samples obtained through interaction with the environment. In our framework, for a given action, the DNN surrogate deterministically or near-deterministically outputs the corresponding performance indicators, thereby defining the next state.

Reward function. The reward function provides the scalar feedback signal that guides learning. Because this study simultaneously minimizes EUIt while maximizing EG and H, a normalized weighted-distance formulation is adopted. First, the three objectives are mapped to a common [0, 1] interval:

$$\begin{aligned} y_{1,\mathrm{norm}} &= \frac{y_1 - y_{1,\min}}{y_{1,\max} - y_{1,\min}}, \\ y_{2,\mathrm{norm}} &= \frac{y_2 - y_{2,\min}}{y_{2,\max} - y_{2,\min}}, \\ y_{3,\mathrm{norm}} &= \frac{y_3 - y_{3,\min}}{y_{3,\max} - y_{3,\min}}. \end{aligned} \tag{4}$$

In the normalized space, the utopian point is defined as [0, 1, 1], representing the ideal state in which all objectives simultaneously reach their best values. The weighted Euclidean distance between the current state and the utopian point is then calculated as

$$d_{\mathrm{weighted}} = \sqrt{(w_1 y_{1,\mathrm{norm}} - 0)^2 + (w_2 y_{2,\mathrm{norm}} - 1)^2 + (w_3 y_{3,\mathrm{norm}} - 1)^2}, \tag{5}$$

where w_1, w_2, and w_3 are preset non-negative weight coefficients whose sum equals 1. In this study, the normalization bounds are derived from the sampled design domain represented in the surrogate-training dataset. This choice makes the reward scale dependent on dataset coverage, and therefore the bound selection must be interpreted as a practical normalization strategy rather than a theoretically unique definition. Finally, the reward function is defined as

$$R = 10^6 - d_{\mathrm{weighted}}. \tag{6}$$

The reward function therefore transforms the multi-objective design problem into a scalar optimization target that still preserves design preferences through the weight coefficients. The factor 10^6 is used only as a numerical offset to keep rewards positive and to stabilize optimization logs; it does not alter the ranking induced by the weighted distance.

2.4.2. DDPG Agent Implementation

To solve the formulated MDP, this study adopts the DDPG algorithm. DDPG is a model-free actor-critic method well suited to high-dimensional continuous action spaces. It concurrently trains an actor network, which selects actions, and a critic network, which evaluates the value of those actions under the reward function. Here, the focus is on the implementation used for urban morphology optimization, while additional convergence and benchmark details are reported in Appendix B.

The implementation is structured around a multi-scenario analysis designed to investigate the agent's ability to generate targeted design solutions based on varying decision-maker preferences. Multiple independent DDPG agents are trained, each guided by a unique reward function reflecting a specific design priority. The common architectural hyperparameters are summarized in Table 5.

Table 5: Hyperparameter settings for the DDPG agent.

Parameter	Value
Reward discount factor (γ)	0.999
Learning rate (actor)	0.001
Learning rate (critic)	0.002
Layers (actor network)	2
Units per layer (actor)	64, 32
Layers (critic network)	2
Units per layer (critic)	64, 32
Soft update factor (τ)	0.001
Batch size	128
Replay buffer size	1×10^6
Maximum episodes	600
Max steps per episode	40
Noise decay factor	0.9998

The actor and critic networks both use two hidden layers with 64 and 32 neurons. This relatively simple architecture was found to be effective for the problem complexity while avoiding the instability often associated with deeper networks. The critic learning rate is slightly larger than the actor learning rate, allowing the value function to stabilize faster and provide a reliable training signal to the policy network. The discount factor is set to 0.999, encouraging farsighted strategies focused on long-term cumulative rewards.

To ensure adequate exploration, Gaussian noise with an initial variance of 1.0 is added to the actor outputs and gradually reduced using a decay factor of 0.9998. A replay buffer with a capacity of one million transitions is used to decorrelate samples and stabilize learning; at each training step, a mini-batch of 128 transitions is sampled to update the networks.

The central pillar of the implementation is the configuration of three distinct optimization scenarios. These scenarios are realized by adjusting the weight coefficients (w_1, w_2, w_3) in the reward function:

- **Scenario 1: Balanced performance.** Equal weights are assigned, $w_1 = w_2 = w_3 = \frac{1}{3}$, producing an unbiased preference among minimizing EUIt, maximizing EG, and maximizing H.
- **Scenario 2: Energy saving focus.** The weights are set to $(w_1, w_2, w_3) = (0.6, 0.2, 0.2)$, prioritizing reduced energy consumption.
- **Scenario 3: Energy generation focus.** The weights are set to $(w_1, w_2, w_3) = (0.2, 0.6, 0.2)$, prioritizing renewable energy production.

By training a separate agent for each of these scenarios, the DRL framework is transformed from a single-solution optimizer into a flexible decision-support tool that supports a priori articulation of design intent. The overall l. . . 404 tokens truncated. . . ons remain tightly concentrated around the one-to-one line for all three objectives, while the residual distributions in Fig. 5 indicate that most prediction errors remain within the $\pm 2\sigma$ band. Even so, these error statistics should be interpreted together with the extreme-region diagnostics and candidate re-evaluation results, because optimization concentrates attention on the parts of the response surface where surrogate misspecification is most consequential.

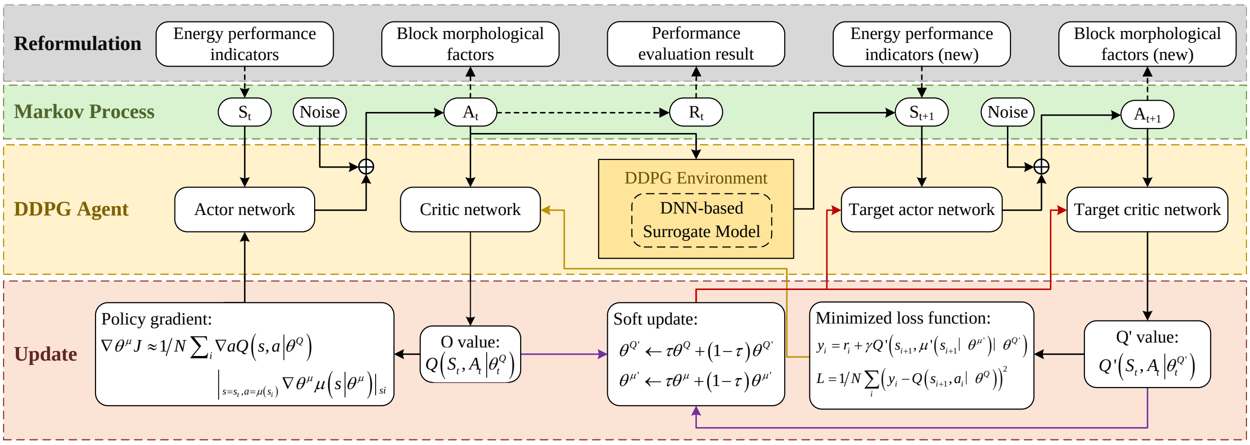

Figure 3: Architecture of the DDPG agent, showing the actor-critic structure and learning process.

3. Results

3.1. Surrogate Model Performance

Based on the simulations generated through the Grasshopper process, a surrogate scale study was carried out over 500, 1000, 1500, and 2000 simulated morphologies in order to identify the highest-accuracy model available to this project. Under the strict accuracy-first rule adopted in this revision, the final optimizer comparison uses the 2000-sample `tuned_standard` surrogate, which applies standard scaling to both inputs and targets and trains a single-hidden-layer network with 128 units. To avoid common issues such as overfitting and training instability, early stopping and dropout techniques were employed throughout the candidate search.

As summarized in Tables 3 and 4, the final DNN achieved high predictive accuracy and robust generalization on the expanded 2000-sample regime. Appendix A further reports RMSE values of 0.279 for EUIt, 0.033 for EG, and 0.032 for H, with normalized RMSE values below 3.1% for all three targets. The parity plots in Fig. 4 show that the predictions remain tightly concentrated around the one-to-one line, while the residual distributions in Fig. 5 indicate that most prediction errors remain within the $\pm 2\sigma$ band. These statistics should nevertheless be interpreted together with the extreme-region diagnostics in Appendix A and the candidate re-evaluation results in Appendix B, because optimization is most sensitive to misspecification in the parts of the response surface that attract the search process.

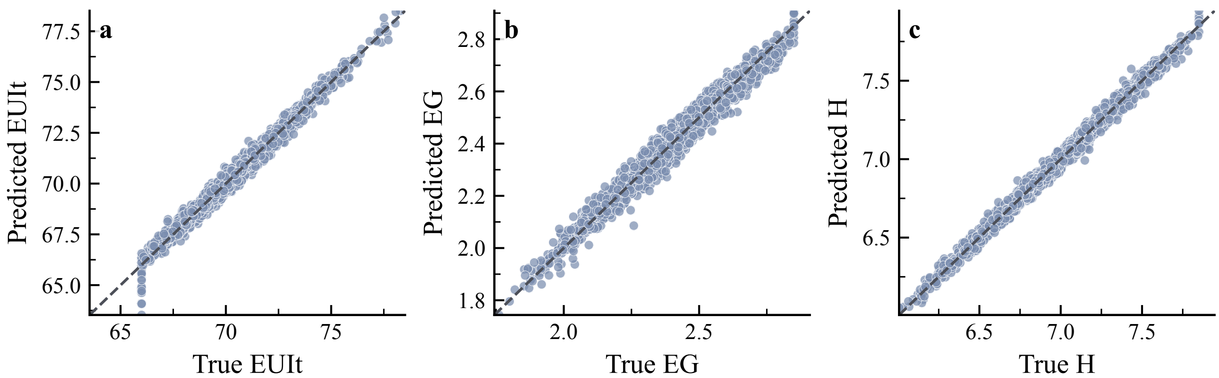

Figure 4: Parity plots comparing surrogate predictions against simulated targets for the three objectives: (a) EUIt, (b) EG, and (c) H. The dashed line denotes perfect agreement.

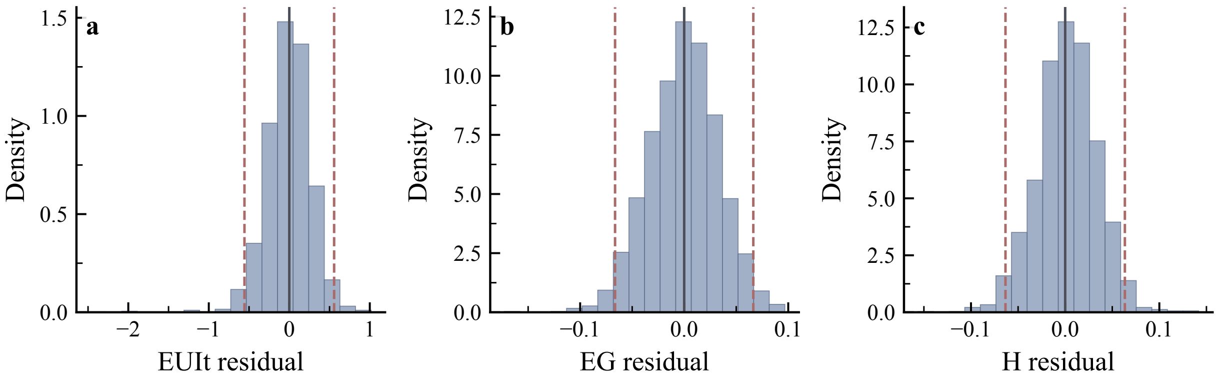

Figure 5: Residual distributions of the surrogate predictions for the three objectives: (a) EUIt, (b) EG, and (c) H. Dashed lines indicate the $\pm 2\sigma$ range around zero residual.

3.2. DDPG Training Dynamics

The learning dynamics of the agents were monitored to assess whether the policy updates approached a stable regime. Figure 6 summarizes the mean learning curves and one-standard-deviation envelopes across the available seeds for the three DDPG scenarios. In the revised server-side reruns, each scenario was trained with 20 seeds and each seed was run for 600 episodes under the guarded surrogate evaluator, which provides a more informative view of training behavior than the earlier single-curve presentation.

The multi-seed curves show that cumulative reward increases sharply during training and then transitions into a narrower high-reward regime, indicating a shift from broad exploration to more concentrated surrogate-region search. On the highest-accuracy 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401, 320, and 398 for the balanced, saving-focused, and generation-focused scenarios, respectively. Relative to the earlier lower-scale rerun, the balanced scenario is visibly smoother and exhibits lower cross-seed fluctuation in reward and output trajectories, whereas the saving-focused and generation-focused scenarios do not become uniformly less variable across all metrics. At the same time, late-stage regression remains common, with mean best-to-final reward gap ratios around 0.62, 0.71, and 0.71 and substantial-regression fractions of 95%, 85%, and 100% across the three scenarios. These curves should therefore be interpreted as evidence that policy learning can reach high-performing surrogate regions rather than as proof of stable asymptotic convergence or global optimality. The learning-curve evidence is therefore complemented by benchmark comparisons and independent candidate re-evaluation.

The three scenarios remain distinguishable but still overlap in aggregate objective space. In the 20-seed reruns on the highest-accuracy surrogate, the energy-saving-focused scenario yields the lowest mean EUIt (67.10 kWh/m^2/y), the generation-focused scenario yields the highest mean EG (2.763×10^6 kWh/y), and the balanced scenario retains the highest mean H (7.687 h). These differences are more visible than in the earlier lower-scale comparison, but they should still be interpreted as preference-conditioned shifts within one surrogate landscape rather than as proof of sharply separated optima.

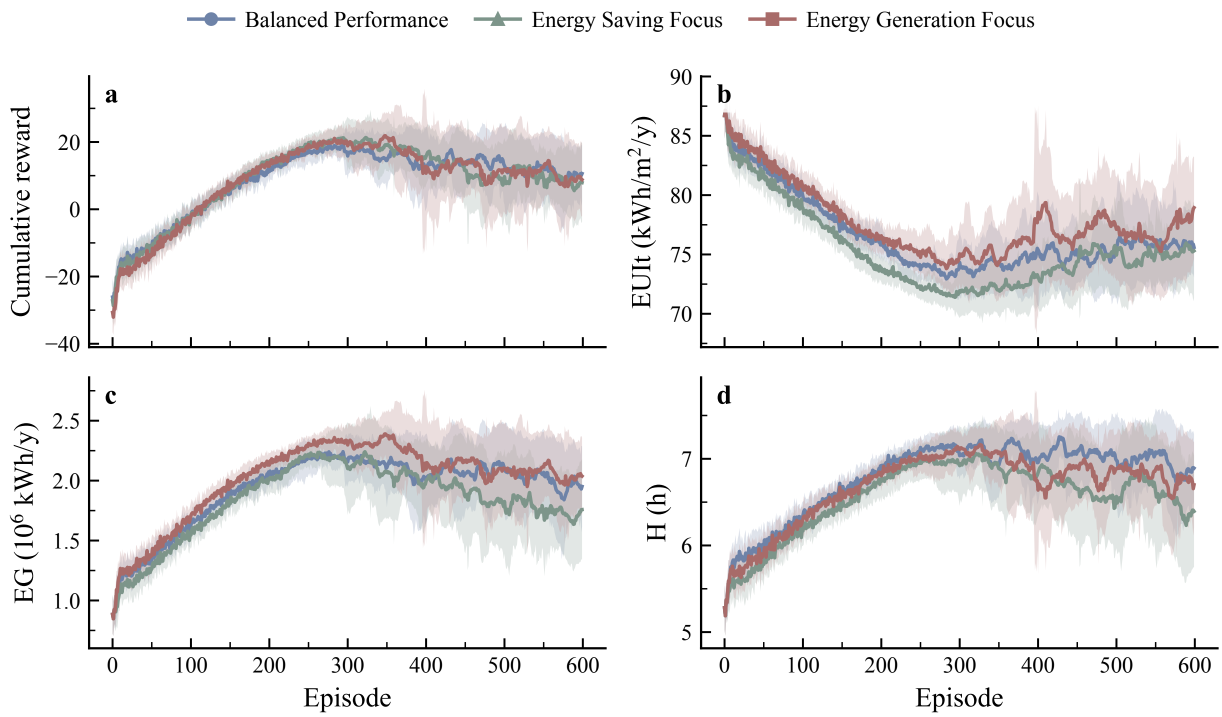

Figure 6: DDPG learning curves during guarded multi-seed training. Solid lines denote mean trajectories across 20 seeds per scenario and shaded regions indicate one standard deviation.

3.3. Multi-objective Optimization Results

Before presenting the comparative results, it is necessary to clarify how the DRL solution clusters were generated. To ensure statistical robustness, each of the three DRL scenarios was trained independently across 20 random seeds. This produces a distribution of high-performance solutions for each agent, reflecting the stable convergence region of its learned policy. The analysis below first explores the learned tendencies of the three DRL scenarios and then compares them against the NSGA-II benchmark.

3.3.1. Multi-scenario DRL Behavior

The primary strength of the DRL framework lies in its flexibility to generate targeted design solutions. By specifying a design preference a priori through reward-function weights, the framework trains three reward-weighted agents under the same surrogate environment. Figure 7 illustrates the differences among the scenarios. In the strict highest-accuracy surrogate rerun, the three DDPG scenarios still occupy overlapping regions, but the scenario-level objective differences are larger than in the earlier matched-checkpoint comparison and remain readable through their morphology summaries.

The most important observation is that the reward-conditioned DRL agents still identify coherent preference-conditioned surrogate regions, but they do not surpass the fair-budget NSGA-II archive on the highest-accuracy surrogate. As shown in Fig. 7, the three DDPG scenarios form overlapping clusters, whereas NSGA-II reaches better average objective values and stronger Pareto indicators on the same 2000-sample surrogate checkpoint. This supports the narrower claim that policy learning can identify a coherent surrogate region, but it also shows that the tested reward-weighted DDPG formulation is not competitive with fair-budget NSGA-II when the comparison is anchored to the strict highest-accuracy surrogate.

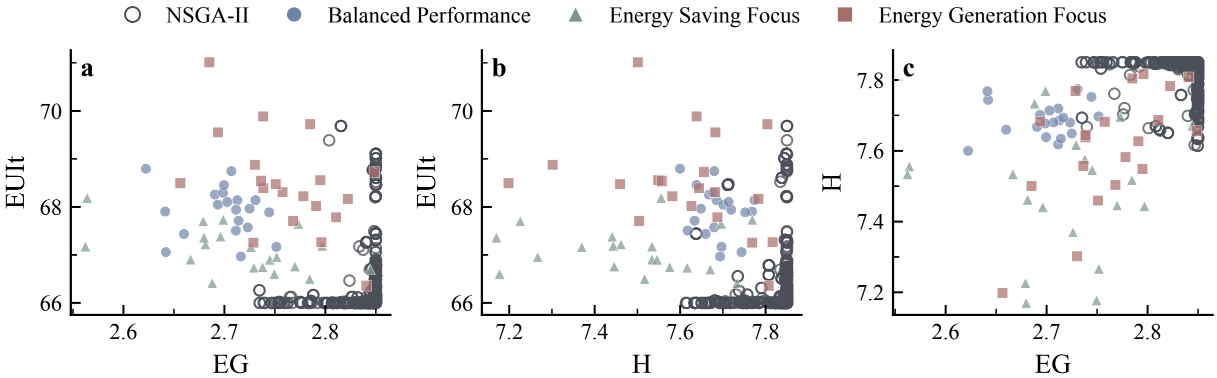

Figure 7: Pairwise objective-space comparison on the strict highest-accuracy 2000-sample surrogate. Each point is one candidate solution. The three DDPG scenarios still form overlapping clusters, while the fair-budget NSGA-II archive reaches stronger objective values on the same selected surrogate.

These objective-space differences become more interpretable in the morphology signatures shown in Fig. 8. The heatmap summarizes relative median UMF levels for each method. In the highest-accuracy surrogate rerun, the balanced scenario still tends toward higher FAR and AF, the generation-focused scenario shifts toward higher SD, OSR, and OSLI, and the saving-focused scenario shows lower EUIt together with slightly lower BD and higher SC. The contrasts are still not dramatic enough to imply cleanly separated optima, but they indicate that the DRL agents produce readable morphology-level fingerprints that vary with the reward specification within the current surrogate workflow.

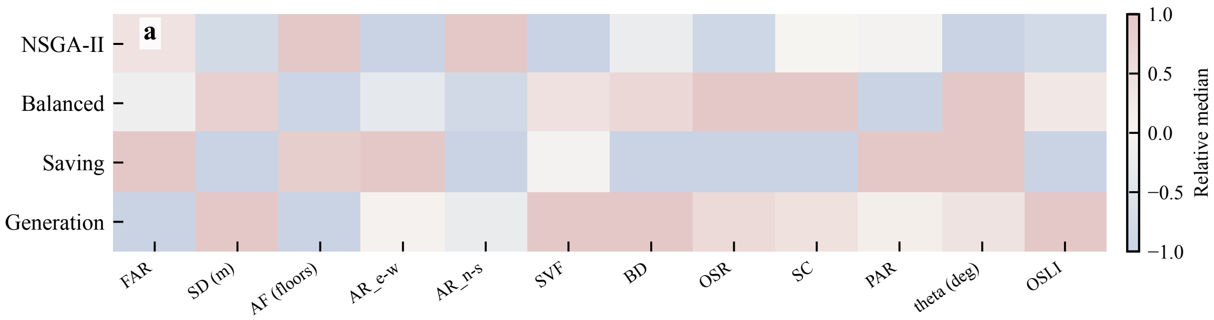

Figure 8: Relative median morphology signatures across methods. Colors encode within-factor contrast across the four method-level median profiles, highlighting which UMFs shift most under the three DDPG reward settings relative to NSGA-II.

3.3.2. *Comparison with NSGA-II*

After establishing the internal behavior of the DRL agents, we compare them to the NSGA-II benchmark on the strict highest-accuracy surrogate selected from the scale study. Table 6 summarizes this apples-to-apples comparison. On this accuracy-first result set, NSGA-II again attains the strongest Pareto-quality metrics (HV = 1.331, IGD = 0.107), while the three DDPG scenarios remain weaker. The generation-focused DDPG run is the strongest of the three DDPG variants in Pareto-quality terms (HV = 1.011, IGD = 0.220), but it still trails NSGA-II. Figure 9 visualizes the same ordering from two complementary angles: scenario-wise objective summaries and preference-aligned utility.

This result changes the interpretation of the benchmark comparison. Rather than claiming superiority of DRL over NSGA-II, the current evidence now supports a narrower conclusion: even after upgrading the surrogate to the strict highest-accuracy 2000-sample model, DDPG does not match fair-budget NSGA-II. Under post-hoc preference-aligned utility selection, the balanced, saving-focused, and generation-focused DDPG candidates all remain below the corresponding best NSGA-II candidate, with utility deltas of approximately −0.267, −0.140, and −0.046, respectively. This conclusion should be interpreted together with three additional caveats. First, the independent re-evaluation reported in Appendix B has not yet been rerun on the 2000-sample comparison, so the latest available candidate-drift evidence still comes from the earlier matched-checkpoint run. Second, the DDPG runs continue to exhibit frequent late-stage reward regression despite reaching high-reward regions mid-training. Third, the optimizer ranking remains checkpoint-sensitive, which means benchmark conclusions should be tied explicitly to the selected surrogate artifact rather than generalized across checkpoints.

Detailed linear correlation maps, joint distributions, and nonlinear response diagnostics are reported in the Supplementary Material rather than in the main text. In the present revision, these diagnostics are used only as interpretive support for the morphology-level shifts in Fig. 8; they are not treated as primary evidence for the benchmark comparison.

Table 6: Accuracy-first comparison across fair-budget NSGA-II and the three DDPG scenarios on the strict highest-accuracy 2000-sample surrogate.

Method	HV	IGD	EUIt	EG (10^6 kWh/y)	H (h)
NSGA-II	1.331	0.107	66.15	2.844	7.836
Balanced DDPG	0.347	0.720	67.90	2.700	7.687
Saving DDPG	0.625	0.549	67.10	2.716	7.487
Generation DDPG	1.011	0.220	68.46	2.763	7.612

Note: The values in this table are computed on the strict highest-accuracy 2000-sample surrogate selected from the scale study. On this surrogate, fair-budget NSGA-II remains stronger than all three DDPG scenarios in both archive-level Pareto metrics and preference-aligned utility.

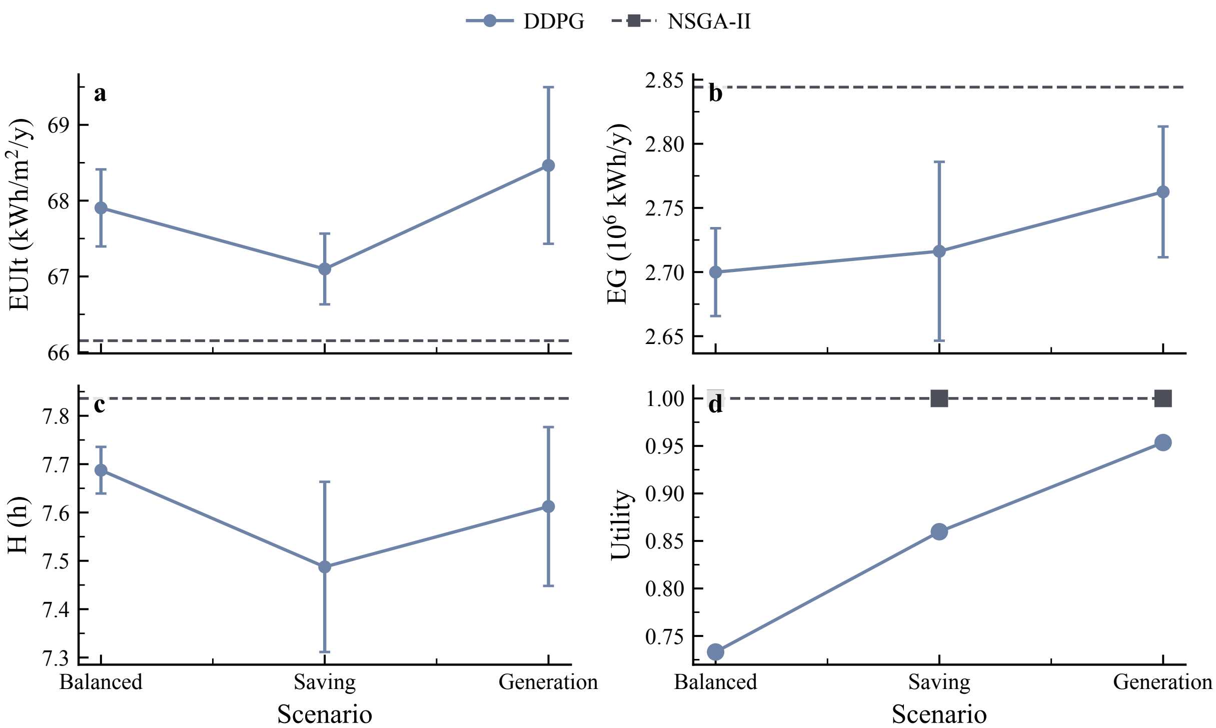

Figure 9: Accuracy-first benchmark summary on the strict highest-accuracy 2000-sample surrogate. Panels (a)–(c) compare scenario-wise objective summaries for DDPG, with the NSGA-II mean shown as a dashed reference line. Panel (d) compares preference-aligned utility for DDPG and NSGA-II across the three scenarios.

Figure 10 summarizes the seed-level convergence diagnostics of the strict highest-accuracy rerun. The generation-focused scenario reaches the highest mean best reward, whereas the saving-focused scenario reaches its plateau earliest. However, all three scenarios still show large best-to-final reward gaps and high late-regression fractions, which confirms that the main role of DDPG in the present workflow is region-finding rather than stable monotonic improvement to a final fixed point.

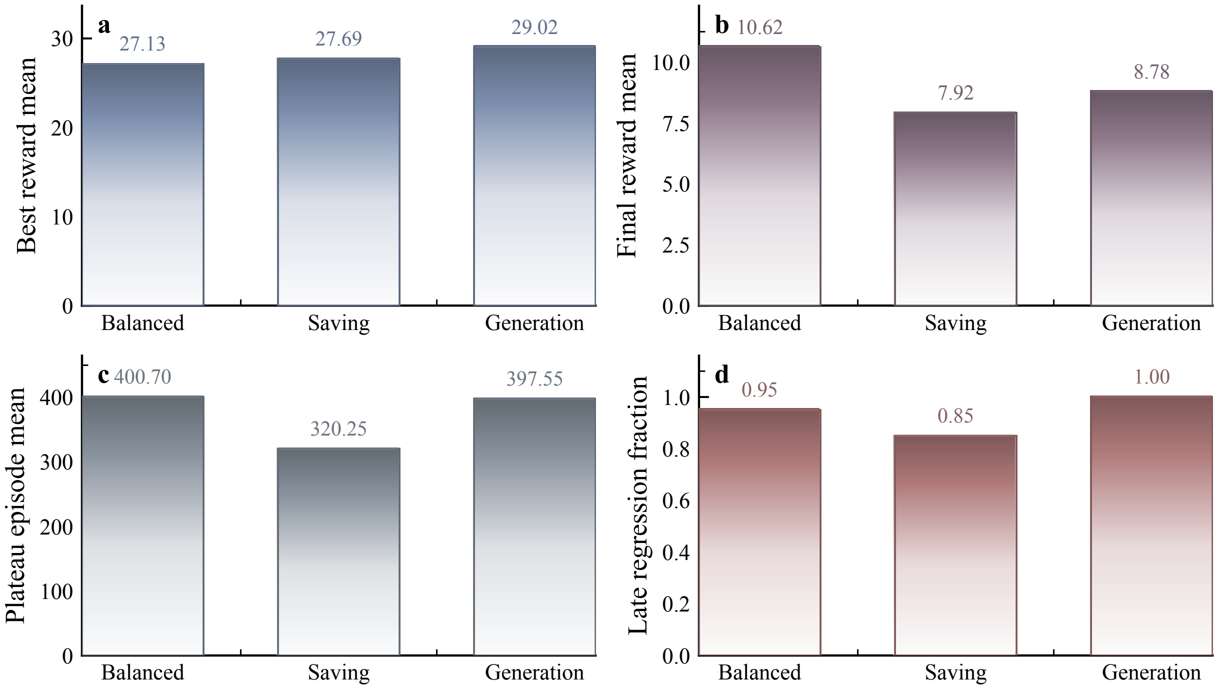

Figure 10: Seed-level convergence diagnostics on the strict highest-accuracy 2000-sample surrogate. Panels report the mean best reward, mean final reward, mean plateau episode, and late-regression fraction across 20 seeds per DDPG scenario.

Figure 11 links the optimizer comparison back to the surrogate-selection procedure itself. The regime-winner diagnostics show that surrogate fidelity improves steadily as the dataset expands from 500 to 2000 samples, and that the strict highest-accuracy rule therefore selects the 2000-sample `tuned_standard` model rather than the smaller 1500-sample alternative that was previously acceptable under a tolerance-based rule. This figure is included to make the paper's final optimizer benchmark traceable to an explicit surrogate-selection decision rather than to an informal checkpoint preference.

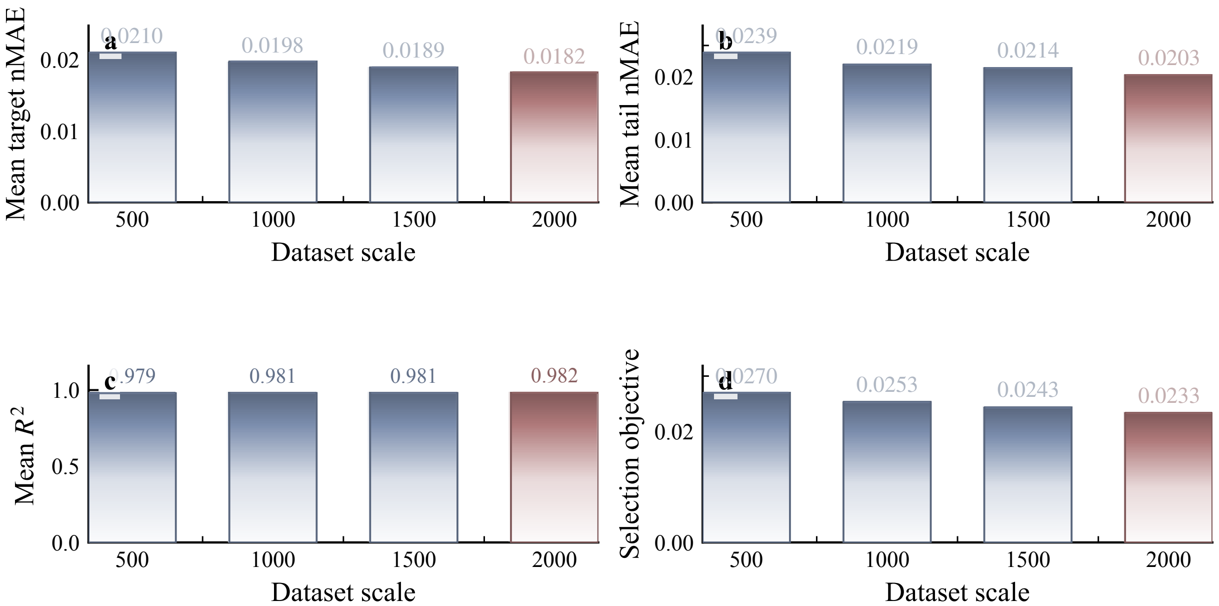

Figure 11: Surrogate scale-study diagnostics across the best regime winner at each dataset scale. Panels report mean target nMAE, mean tail nMAE, mean R^2, and the selection objective from 500 to 2000 samples, with the 2000-sample winner highlighted as the strict highest-accuracy choice.

4. Discussion

This study develops a DRL-DNN integrated framework for block-scale urban morphology design and re-examines its comparative performance under an accuracy-first surrogate-selection protocol. The revised analysis indicates that the framework is most credible as a preference-conditioned design-search tool and as a methodological case study of surrogate-dependent benchmarking, rather than as a superior replacement for Pareto-based evolutionary optimization.

The main methodological finding is that reward-conditioned policy learning can guide the search toward compact high-performing regions on a given surrogate checkpoint, but that advantage is less stable and less competitive than earlier revisions implied. On the strict highest-accuracy 2000-sample surrogate, fair-budget NSGA-II is clearly stronger than all three DDPG scenarios in both objective summaries and Pareto-quality metrics, and it also dominates the post-hoc preference-aligned utility comparison. The three DDPG scenarios remain distinguishable but overlapping in objective space, which suggests that the value of DRL in the present setting lies more in region-finding and interpretable morphology conditioning than in producing clearly separated preference optima.

Even so, the multi-scenario design underscores the flexibility of the DRL framework as a decision-support tool. Unlike NSGA-II, which returns a broad non-dominated archive, the proposed workflow allows designers to encode preferences in advance. In the highest-accuracy rerun, the resulting differences are clearer in morphology descriptors than in top-line objective means, but they can still be interpreted in terms of density, openness, sky access, and orientation.

The framework also provides interpretable design guidance. The correlation analysis and the additional nonlinear diagnostics indicate that openness-related variables such as OSR and SVF, together with density-related variables such as FAR and BD, play a dominant role in shaping the objective responses. Thus, even though DDPG does not outperform NSGA-II on the strict highest-accuracy surrogate, the DRL formulation remains useful for generating policy-conditioned and transferable design heuristics.

Independent candidate re-evaluation further sharpens this interpretation. On the 2000-sample surrogate rerun, the three DDPG winners do not all drift in the same direction: the balanced candidate improves in EUIt but loses EG and

H, whereas the saving-focused and generation-focused candidates drift upward in EUIt and downward in EG after reevaluation through the independent fallback simulation path. The selected NSGA-II candidate also drifts away from its surrogate optimum, but it remains competitive in aggregate objective value. This means the latest results are more balanced than the earlier checkpoint-specific reevaluation, yet they still reinforce the need to keep optimizer claims bounded by surrogate reliability rather than by surrogate-space optimum values alone.

Several limitations should be acknowledged. First, the findings are derived from a single case study in one climate zone. Second, even after expansion to 2000 simulated cases, the surrogate still approximates a 12-dimensional continuous design space and therefore remains vulnerable in sparse or extreme regions; accordingly, the current Pareto claims should be interpreted as applying to the interpolated surrogate region rather than to unconstrained extrapolation beyond the sampled design domain. Third, the current comparison remains surrogate-based during optimization, and the independent re-evaluation still relies on a fallback analytic simulator rather than a full physical stack or external empirical validation dataset. Fourth, the updated reevaluation shows that both DDPG and NSGA-II candidates can drift materially relative to their surrogate predictions, so surrogate-space ranking should not be overgeneralized. Fifth, although the new 2000-sample rerun closes the per-seed and seed-count symmetry between DDPG and NSGA-II, broader optimizer families still need to be tested before claiming general policy-learning advantages. Sixth, the present RL formulation is a scalarized static black-box optimization problem recast as policy learning; therefore, alternative continuous optimizers remain important baselines for future work.

5. Conclusions

This study presents a DRL-DNN framework for co-optimizing urban morphology and energy performance in a surrogate-based setting. The revised evidence suggests that the method should be interpreted as a preference-conditioned policy-learning framework rather than as a universally dominant alternative to Pareto-based search.

Three main conclusions can be drawn. First, the surrogate scale study shows that the strict highest-accuracy model in this project is the 2000-sample `tuned_standard` DNN, which improves mean target nMAE by about 9.9% relative to the best 500-sample baseline. Second, when the optimizer comparison is rerun on that highest-accuracy surrogate and kept within the paper's original three-scenario DDPG-versus-NSGA-II architecture, fair-budget NSGA-II outperforms all three DDPG scenarios in archive-level objective summaries, Hypervolume, IGD, and preference-aligned utility. Third, the DRL scenarios still yield interpretable design fingerprints, particularly through differences in FAR, BD, OSR, SVF, and OSLI, which can be explained through the correlation and joint-distribution analyses.

In summary, this paper transforms the optimization problem from a static black-box search into a policy-learning formulation that can encode design preferences directly. The framework is best viewed as a flexible and interpretable design-search tool whose claims must remain bounded by surrogate reliability, benchmark fairness, and the absence of external empirical validation. Future work should expand the dataset, use broader baselines, and validate the final solutions through direct simulation and external data.

Appendix A. Additional Case Settings and Surrogate Validation

Appendix A.1. Case-setting details

The base urban block is represented as a 240 m × 240 m square subdivided into a 3 × 3 grid of nine land units. Each land unit measures 80 m × 80 m. A two-step inward offset is applied to represent the road boundary and building setback, resulting in a nominal buildable area of 60 m × 60 m for each unit. One land unit is designated as open space, and the remaining units are populated with residential building prototypes.

To avoid notation drift between the main text and the appendix, the following definitions are used consistently:

- **AF** (average floors) is defined as the total gross floor area divided by the total building footprint area.
- **OSR** (open space ratio) is defined as the non-built-up area of the block divided by the total gross floor area.

Table A.7: Representative building prototypes used in the parametric dataset.

ID	Type	Footprint Area (m^2)	Floors	FAR Range
P-1	Point	2000	1–3	0.31–0.94
P-2	Point	1500	4–12	0.94–2.81
P-3	Point	1000	13–30	2.03–4.69
P-4	Point	1000	13–30	2.03–4.69
S-1	Slab	2000	1–3	0.31–0.94
S-2	Slab	1500	4–12	0.94–2.81
S-3	Slab	1000	13–30	2.03–4.69
C-1	Courtyard	2000	1–3	0.31–0.94
C-2	Courtyard	1500	4–12	0.94–2.81

Appendix A.2. Surrogate validation

The original surrogate workflow began from a 500-sample baseline, but the final optimizer comparison in this revision uses the strict highest-accuracy surrogate obtained from the expanded 2000-sample regime. Five-fold cross-validation is used to assess that final selected model. Table A.8 reports the corresponding aggregate validation results.

Table A.8: Five-fold cross-validation summary for the DNN surrogate model.

Objective	Mean R^2	Mean MAE	Std. R^2	Std. MAE
EUIt	0.983	0.209	±0.001	±0.005
EG	0.972	0.026	±0.002	±0.001
H	0.992	0.025	±0.001	±0.001

To evaluate reliability in optimization-relevant regions, additional error slices were computed over the lower and upper tails of each target distribution. The corresponding mean absolute errors are listed in Table A.9. Table A.10 reports complementary surrogate-fidelity statistics from the cross-validation predictions.

Table A.9: Error diagnostics in extreme regions of the target distributions.

Target	$\mathrm{MAE}_{\mathrm{all}}$	MAE_{q10}	MAE_{q90}	$q_{0.10}$	$q_{0.90}$
EUIt	0.209	0.283	0.194	67.83	73.32
EG	0.026	0.027	0.030	2.132	2.655
H	0.025	0.026	0.029	6.514	7.429

Table A.10: Additional surrogate-fidelity audit on the cross-validation predictions of the selected 2000-sample surrogate.

Target	RMSE	MAE	nRMSE	Tail nMAE	R^2
EUIt	0.279	0.209	0.0227	0.0194	0.983
EG	0.033	0.026	0.0305	0.0265	0.972
H	0.032	0.025	0.0172	0.0150	0.992

Note: nRMSE is normalized by the observed range of each target over the cross-validation predictions. Tail nMAE is the mean of the lower-tail and upper-tail nMAE diagnostics used for surrogate selection.

Appendix B. Benchmark Robustness and Independent Re-evaluation

Appendix B.1. Training dynamics and benchmark protocol

The revised server-side reruns used for the final accuracy-first comparison were executed under the shared guarded surrogate evaluator to suppress exploitation of extrapolative edge cases. On the strict highest-accuracy 2000-sample surrogate, seed-level diagnostics show that the 20-episode rolling reward typically reaches 95% of its best observed value around episodes 401 (balanced), 320 (saving), and 398 (generation). Compared with the earlier lower-scale rerun, the balanced scenario becomes visibly smoother and less noisy across seeds, but the saving-focused and generation-focused scenarios do not show a uniform reduction in cross-seed variability across reward and objective trajectories. However, late-stage regression remains common: the mean best-to-final reward gap ratios are about 0.62, 0.71, and 0.71 for the balanced, saving, and generation scenarios, respectively, and the substantial-regression fractions remain high at 95%, 85%, and 100%.

For benchmark comparison, the revised implementation removes the previous benchmark-matching shortcut from the publication-mode NSGA-II path. Publication mode compares DDPG and NSGA-II under a shared surrogate environment, matched per-seed evaluation budget, and matched seed count, and applies a post-hoc preference-aligned utility summary consistently across methods. On the strict highest-accuracy 2000-sample surrogate, NSGA-II remains stronger than all three DDPG scenarios. The corresponding budget accounting and objective-space separation are summarized in Table B.11.

Table B.11: Budget accounting and objective-space separation in the highest-accuracy 2000-sample rerun.

Item	Value 1	Value 2	Interpretation
DDPG per-seed budget	600 episodes	40 steps/episode	24,000 surrogate interactions
DDPG seeds per scenario	20	–	pooled into one scenario archive
NSGA-II seeds in rerun	20	–	pooled into one comparison archive
Distance: balanced vs generation	0.5417	–	normalized objective-space distance of scenario means
Distance: balanced vs saving	0.6809	–	normalized objective-space distance of scenario means
Distance: generation vs saving	0.7619	–	normalized objective-space distance of scenario means
NSGA-II archive spread	0.1642	0.1968	mean and std of distance to normalized archive center

Note: The 2000-sample highest-accuracy rerun matches both the per-seed interaction budget and the number of seeds between DDPG and NSGA-II. The DDPG scenarios are more separated than in the earlier matched-checkpoint run, but NSGA-II still dominates the preference-aligned utility comparison.

To complement the linear correlation results in the main text, Fig. B.12 reports representative surrogate-based nonlinear response profiles for OSR→EUIt, FAR→EG, SVF→H, and θ →H.

Nonlinear response profile figure not available in the current worktree. The text and tables in this draft already reflect the updated highest-accuracy-surrogate rerun.

Figure B.12: Surrogate-based nonlinear response profiles for selected UMF-target pairs: (a) OSR→EUIt, (b) FAR→EG, (c) SVF→H, and (d) θ →H.

Appendix B.2. Independent re-evaluation of selected candidates

To reduce over-reliance on surrogate predictions, the workflow re-evaluates preference-aligned candidate solutions using an independent simulation call. In the current local environment, this re-evaluation still uses the fallback analytic simulator, but the code path is structured so that a full physical-stack simulation can replace it when server-side results become available. Table B.12 reports the latest available reevaluation results for the strict highest-accuracy 2000-sample comparison. The updated evidence shows heterogeneous drift: the balanced DDPG winner improves in EUIt but loses EG and H, the saving-focused and generation-focused DDPG winners drift upward in EUIt and downward in EG, and the selected NSGA-II candidate also moves away from its surrogate prediction. This narrows the earlier asymmetry between DDPG and NSGA-II, but it still indicates that surrogate-space ranking should not be treated as a direct substitute for independent validation.

Table B.12: Independent re-evaluation of selected candidates from the highest-accuracy 2000-sample surrogate rerun. Surrogate and deterministic re-evaluation values are reported for the current preference-aligned selections.

Method	Scenario	$EUIt_{sur}$	$EUIt_{re}$	EG_{sur}	EG_{re}	H_{sur}	H_{re}
DDPG	Balanced	67.168	66.000	2.751	2.689	7.697	7.374
NSGA-II	NSGA-II (seed 20260310)	66.000	67.730	2.850	2.712	7.850	7.510
DDPG	Saving	66.698	67.473	2.844	2.769	7.671	7.611
DDPG	Generation	66.356	67.836	2.841	2.735	7.807	7.530

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